

NASTARAN DARABI

Engineering Research Facility (ERF) 2024, 842 West Taylor Street, Chicago, IL

✉ ndarab2@uic.edu  [linkedin.com/in/nastaran-darabi](https://www.linkedin.com/in/nastaran-darabi)  github.com/nstrndrbi  [Google Scholar](#) 📞 +1(773) 551-1451

EDUCATION

University of Illinois Chicago

Jan. 2021 - March 2026

Doctor of Philosophy, Electrical and Computer Engineering

Chicago, IL, USA

Thesis: "Ensuring Robustness and Reliability in Multimodal Autonomous Systems:

A Unified Framework for Multi-Level Anomaly Detection, Adaptive Resilience, and Causal Diagnostics"

University of Illinois Chicago

Jan. 2024 - May 2025

Master of Science, Computer Science (Concurrent)

Chicago, IL, USA

Natural Language Processing and Machine Learning. GPA: 4/4

University of Illinois Chicago

Jan.2021 - May 2025

Master of Science, Electrical and Computer Engineering

Chicago, IL, USA

Deep Learning. GPA: 4/4

EXPERIENCE

Apple

May 2025-September 2025

Machine Learning Intern

Cupertino, CA, United States

- Contributed to Apple's "Smart Testing" initiative, focusing on AI-driven process optimization and defect detection for quality control in high-volume manufacturing. "Smart Testing" could save 57.79% cost with 0 escape units.
- Engineered solutions for extremely imbalanced data and integrated algorithms into manufacturing equipment to streamline the development-to-production pipeline, providing real-time analysis to improve efficiency.
- Collaborated with cross-functional teams at Apple to ensure the effective implementation and validation of ML solutions.

Radical AI

April 2024-December 2024

Artificial Intelligence Engineer Intern

Remote, United States

- Utilized cutting-edge technologies, including OpenAI and Google Gemini, to develop ReX, an AI Coach designed to be a reliable career companion for learners. ReX offers personalized coaching, mentorship, and support across different stages of their career life-cycle.

Semiconductor Research Corporation(SRC)

January 2023-Present

Research Scholar

Remote, United States

Efficient multi-modality sensor fusion for 3D object detection: Conducting research on adaptive LiDAR for 3D object detection.

- Publication: "Enhancing 3D Robotic Vision Robustness by Minimizing Adversarial Mutual Information through a Curriculum Training Approach": proposed a method that minimizes both prediction loss and mutual information (MI) under adversarial attacks, using curriculum advisors to gradually introduce adversarial objectives and prevent model overwhelm. This approach improves robustness, achieving 2-5% accuracy gains on ModelNet40 and 5-10% mAP improvements on KITTI, compared to the SOTA.

Knowles Corporation

May 2022 - August 2022

Failure Analysis Intern

Itasca, IL, United States

- Developed and implemented a deep learning algorithm focused on the detection of microphone failures through analysis of X-ray images. I utilized advanced computer vision techniques to enhance the accuracy and efficiency of failure detection processes. Also, I collaborated with a multidisciplinary team to integrate the algorithm into the existing diagnostic framework, ensuring seamless operation.

University of Illinois Chicago

January 2021-Present

Graduate Research Assistant

Chicago, IL, United States

Thesis: "Ensuring Robustness and Reliability in Multimodal Autonomous Systems:

A Unified Framework for Multi-Level Anomaly Detection, Adaptive Resilience, and Causal Diagnostics"

- Selected Publication: Developed "EigenShield", an inference-time defense leveraging Random Matrix Theory to enhance adversarial robustness in vision-language models. In evaluations, EigenShield reduced the attack success rate by 76.5% for LLaVA-v1.5-7B and by 45.3% for MiniGPT-4, while also significantly lowering toxicity scores.

TECHNICAL SKILLS

- **Languages:** Python, Matlab, Perl, SQL, NoSQL, R, C/C++, HTML, JavaScript
- **Machine Learning Libraries:** PyTorch, TensorFlow, Scikit-Learn, Pyro, Pandas
- **Technologies:** Git, Figma, Docker
- **LLM Tools and Frameworks:** Hugging Face Transformers, LangGraph, LlamaIndex, Agents, CrewAI, fedLLM
- **Relevant Courses:** (1) Machine Learning (Intro and Adv.), (2) Computer Vision (Intro and Adv.), (3) Causal Inference, (4) Natural Language Processing (Intro and Adv.), (5) Computer Algorithms, (6) Neural Networks (Intro and Adv.), (7) Reinforcement Learning, (8) Large Language Models in Production, (9) Algorithmic Fairness, (10) Database Systems

RESEARCH INTERESTS

Vision-Language Models, Multimodal Deep Learning, Vision-Language-Action Models, Ambient LLM Agents, Adversarial Machine Learning, Robust AI, Reinforcement Learning, Large Language Models, Causal Inference

SELECTED PUBLICATIONS

- **Nastaran Darabi**, Amit Ranjan Trivedi, “ProGAL-VLA: Grounded Alignment through Prospective Reasoning in Vision-Language-Action Models”, **The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR, 2026)**, Under Review
- **Nastaran Darabi**, Devashri Naik, Sina Tayebati, Dinithi Jayasuriya, Amit Ranjan Trivedi, “Resilience in Ambient Multi-Agent LLMs via Decentralized Bio-Autonomic Control and Immune-Inspired Anomaly Detection”, **Association for the Advancement of Artificial Intelligence Conference (AAAI, 2026)**, Accepted
- **Nastaran Darabi**, Devashri Naik, Sina Tayebati, Dinithi Jayasuriya, Ranganath Krishnan, Amit Ranjan Trivedi, “EigenShield: Causal Subspace Filtering via Random Matrix Theory for Adversarially Robust Vision-Language Models”, **Association for the Advancement of Artificial Intelligence Conference (AAAI, 2026)**, Accepted
- Dinithi Jayasuriya, Divake Kumar, Sureshkumar Senthilkumar, Devashri Naik, **Nastaran Darabi**, Amit Ranjan Trivedi, “Causal-Guided Dimension Reduction for Efficient Pareto Optimization”, **Design, Automation and Test in Europe Conference (DATE, 2026)**, Accepted
- **Nastaran Darabi**, Divake Kumar, Sina Tayebati, Amit Ranjan Trivedi, “INTACT: Inducing Noise Tolerance through Adversarial Curriculum Training for LiDAR-based Safety-Critical Perception and Autonomy”, **IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP, 2026)**, Under Review
- Davide Etti, **Nastaran Darabi**, S. Senthilkumar, Amit R. Trivedi, “RMT-KD: Random Matrix Theoretic Causal Knowledge Distillation”, **Design, Automation and Test in Europe Conference (DATE, 2026)**, Under Review
- Davide Etti, **Nastaran Darabi**, Sina Tayebati, M.Subedar, Ranganath Krishnan, and Amit R. Trivedi, “EigenTracks: Spectral Activation Feature Tracking for Robust and Real-Time Hallucination and OOD Detection in LLMs and VLMs”, **IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP, 2026)**, Under Review
- Sina Tayebati, Divake Kumar, **Nastaran Darabi**, Dinithi Jayasuriya, Ranganath Krishnan, Amit Ranjan Trivedi, “Learning Conformal Abstention Policies for Adaptive Risk Management in Large Language and Vision-Language Models”, **Asian Conference on Machine Learning (ACML, 2025)**
- A.R. Trivedi, S. Tayebati, H. Kumawat, **N. Darabi**, D. Kumar, A.K. Kosta, Y. Venkatesha, D. Jayasuriya, N. Jayasinghe, P. Panda, S. Mukhopadhyay, K. Roy, “Intelligent Sensing-to-Action for Robust Autonomy at the Edge: Opportunities and Challenges”, **Design, Automation and Test in Europe Conference (DATE, 2025)**
- **Nastaran Darabi**, Dinithi Jayasuriya, Devashri Naik, Theja Tulabandhula, and Amit Ranjan Trivedi, “Enhancing 3D Robotic Vision Robustness by Minimizing Adversarial Mutual Information through a Curriculum Training Approach”, **IEEE International Conference on Robotics and Automation (ICRA, 2025)**
- **Nastaran Darabi**, Sina Tayebati, Sureshkumar S., Sathya Ravi, Theja Tulabandhula, and Amit R. Trivedi, “STARNet: Sensor Trustworthiness and Anomaly Recognition via Approximated Likelihood Regret for Robust Edge Autonomy”, **IEEE International Joint Conference on Neural Networks (IJCNN, 2024)**
- **Nastaran Darabi**, Priyesh Shukla, Dinithi Jayasuriya, Divake Kumar, A.C. Stutts, Amit R. Trivedi, “Navigating the Unknown: Uncertainty-Aware Compute-in-Memory Autonomy of Edge Robotics”, **Design, Automation and Test in Europe Conference (DATE, 2024)**
- Domenico Parente, **Nastaran Darabi**, A.C. Stutts, Theja Tulabandhula, and Amit R. Trivedi, “Conformalized Multimodal Uncertainty Regression and Reasoning”, **IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP, 2024)**

SELECTED PROJECTS

- LLM Agent for Quantum Compute Resource Optimization — CS 532 Spring 2025
- Enhancing LLM Safety with Random Matrix Theory — CS 521 Spring 2025
- Enhancing AI Visibility, Fairness, and Accountability — CS 516 Spring 2025
- UIC Course Planner and Catalog: build the front-end of a website Fall 2024
- Design a Specialized Chatbot from Scratch — CS 421 Fall 2024
- Ethereum Blockchain Transaction Analysis — CS 418 Fall 2024
- Point cloud generation techniques using Normalizing Flows and Custom Guidance — CS 518 Fall 2023
- Bridging Causal Inference and Sensor Trustworthiness for Anomaly Recognition — CS 520 Fall 2023
- Liver and Liver-Tumor Segmentation — CS 515 Spring 2023
- Design LSTM model for name generation — CS 559 Fall 2022
- Small Digital Thermometer Summer 2020
- Audio Amplifier Design Summer 2020
- Line Follower Robot Summer 2020
- Small Digital Oscilloscope Summer 2020

CERTIFICATIONS AND AWARDS

- Introduction to Back-End Development — [Credential URL](#) Summer 2025
- Building Transformer-Based Natural Language Processing Applications — [Credential URL](#) Fall 2023
- Supervised Machine Learning: Regression and Classification — [Credential URL](#) Summer 2022
- Advanced Learning Algorithms — [Credential URL](#) Summer 2022
- 2022 College of Engineering Exceptional Teaching Promise Award Spring 2022
- 2021 Complete Python Bootcamp From Zero to Hero in Python — [Credential URL](#) Fall 2021

AFFILIATIONS AND SERVICES IN RESEARCH COMMUNITY

- **Graduate Student Member, IEEE, 2021-present**
- **Graduate Student Member, IEEE Robotics and Automation Society, 2023-present**
- **IEEE Women In Engineering (WIE) Member, 2023-present**
- **Reviewer:** AAAI Conference on Artificial Intelligence, IEEE International Conference on Robotics and Automation (ICRA), Conference on Neural Information Processing Systems (NeurIPS), IEEE Transactions on Circuits and Systems for Video Technology, IEEE Transactions on Artificial Intelligence, IEEE Access, International Joint Conference on Neural Networks (IJCNN), IEEE Transactions on Systems, Man, and Cybernetics: Systems

REFERENCES

- **Dr. Amit Ranjan Trivedi**
*Associate Professor, Department of Electrical and Computer Engineering
University of Illinois at Chicago, Illinois, United States (Email: amitrt@uic.edu)*
- **Dr. Daniela Tuninetti**
*Professor and Department Head, Department of Electrical and Computer Engineering
University of Illinois at Chicago, Illinois, United States (Email: danielat@uic.edu)*
- **Dr. Omesh Tickoo**
*Senior Principal Engineer (AI) and Senior Director at Intel
(Email: omesh.tickoo@intel.com)*
- **Ranganath Krishnan**
*Distinguished ML Engineer and Director, AI Labs at Capital One
(Email: rangnanath.krishnan@capitalone.com)*